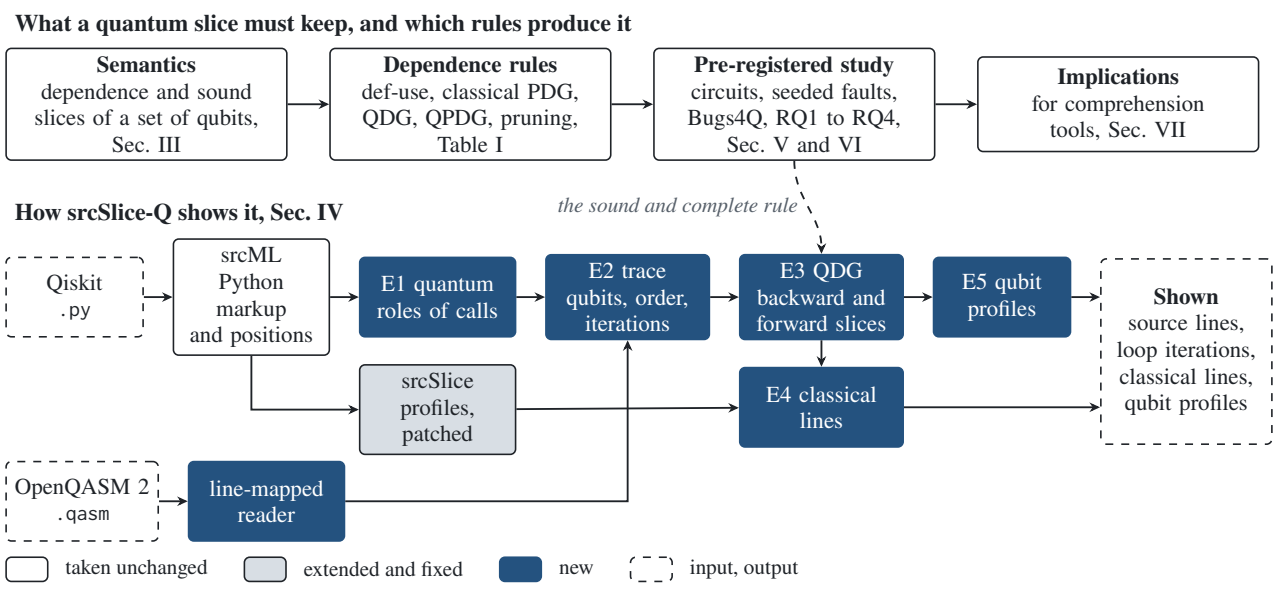

What a quantum slice must keep, and which rules produce it
Semantics
dependence and sound slices of a set of qubits, Sec. III
Dependence rules
def-use, classical PDG, QDG, QPDG, pruning, Table I
Pre-registered study
circuits, seeded faults, Bugs4Q, RQ1 to RQ4, Sec. V and VI
Implications
for comprehension tools, Sec. VII
the sound and complete rule
How srcSlice-Q shows it, Sec. IV
Qiskit .py
srcML Python markup and positions
E1 quantum roles of calls
E2 trace qubits, order, iterations
E3 QDG backward and forward slices
E5 qubit profiles
Shown
source lines, loop iterations, classical lines, qubit profiles
srcSlice profiles, patched
E4 classical lines
OpenQASM 2 .qasm
line-mapped reader
taken unchanged
extended and fixed
new
input, output